LESSON 6.15

Simplify Algebraic Expressions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can simplify algebraic expressions by combining like terms.

WHY THIS MATTERS The music studio is placing supply orders for three rehearsal rooms. Room A needs 4 microphones and 3 stands. Room B needs 2 microphones and 5 stands. Room C needs 6 microphones and 1 stand. Instead of ordering separately, you can combine: (4 + 2 + 6) microphones and (3 + 5 + 1) stands = 12 microphones and 9 stands. That's simplifying by combining like items!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Like Terms *Spanish: Términos semejantes*	Terms with the same variable raised to the same exponent, like 2x and 5x.	5x and 3x are like terms (both x); 5x and 5y are NOT (different variables); 2x and 2x² are NOT (different powers)
Combine *Spanish: Combinar*	To add or subtract terms with the same variable and exponent into one.	8m + 3m = 11m — add the coefficients (8 + 3 = 11) and keep the variable (m)
Simplify *Spanish: Simplificar*	To rewrite an expression in a shorter form that is still equal to the original.	2a + 5a + 3 simplifies to 7a + 3 — the two a-terms combine, the constant stays
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	In 9x, the coefficient is 9 — it tells you 'nine groups of x'; if x = 2, then 9x = 18
Term *Spanish: Término*	A single number, variable, or product of them, separated by + or − signs.	In 3x + 5 − 2x, the terms are 3x, 5, and 2x.
Distributive property *Spanish: Propiedad distributiva*	Multiplying a number by everything inside the parentheses: a(b + c) = ab + ac.	3(x + 4) = 3·x + 3·4 = 3x + 12.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

The rooms need microphones and stands: (4 + 2 + 6) microphones and (3 + 5 + 1) stands.

1. Microphones are like items, so I add: 4 + 2 + 6 = 12 microphones.
2. Stands are like items, so I add: 3 + 5 + 1 = 9 stands.
3. In algebra this is the same as 4m + 2m + 6m + 3s + 5s + 1s = 12m + 9s.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's simplify 5x + 3x. Are these like terms?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Two terms are like terms when they have the same _____. You cannot combine _____ and _____ because they represent different _____.

Word bank Like Terms | Combine | Simplify | Coefficient | Term | Distributive property | variable | like terms | coefficient | combine | same variable | different

WATCH OUT!

A common mistake

A common mistake in Simplify Algebraic Expressions is sweeping a constant into the variable terms

LESSON 6.15

Simplify Algebraic Expressions

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Simplify 7x + 3x.

(A) 10x
(B) $10x^2$
(C) 21x
(D) 73x

STEPS TO TRY

1. 7x and 3x have the same variable x, so they're like terms and can combine.
2. Add the coefficients; the x stays exactly the same — no squaring, no multiplying coefficients.
3. Compute 7 + 3 and place that sum in front of x.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Simplify $9n + 4 + 2n$.

(A) $11n + 4$

(B) $11n^2 + 4$

(C) $15n$

(D) $92n + 4$

STEPS TO TRY

1. There are two n-terms (9n and 2n) and one lonely constant (4) with no partner.
2. Combine only the like terms; the constant 4 stays separate in your answer.
3. Add 9n + 2n by adding 9 + 2. The 4 keeps its own spot at the end.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which pair are like terms?

(A) $6x$ and $2x$

(B) $6x$ and $6y$

(C) $6x$ and 6

(D) $6x$ and $2x^2$

STEPS TO TRY

1. Like terms need the exact same variable raised to the exact same power.
2. Compare two things in each pair: the letter itself and the small exponent on it. Both must match exactly.
3. Check each pair in order: same variable, and the same exponent? Both must match.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Simplify 5a + 3 + 2a.

(A) 7a + 3

(B) $7a^2 + 3$

(C) 10a

(D) 52a + 3

STEPS TO TRY

1. Spot the like terms: 5a and 2a share the variable a. The 3 has no variable.
2. Combine the a-terms by adding their coefficients; carry the 3 along unchanged.
3. Add 5 + 2 for the a coefficient, then write the constant 3 after it.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What rule determines whether two terms are 'like terms'? Why can't you combine unlike terms?

SENTENCE STARTER Two terms are like terms when they have the same ___. You cannot combine ___ and ___ because they represent different ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help